

02 · Does the offer work better for some segments? — segment effects (pathmc)

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The business decision. The email lifts spend *on average* (that’s the ATE from notebook 01). But an average can hide everything that matters: maybe the email really moves **high-value, highly-engaged** customers and barely touches the rest. If the segments genuinely differ, we should write segment-specific send rules; if the apparent “gap” is just noise, chasing it would be overfitting our marketing to randomness. Either way we want to know **what it costs to treat everyone as the average**.

1.0.1 The concept: moderation and the CATE

When a treatment’s effect *depends on who receives it*, we say the effect is **moderated** by the customer’s characteristics, and the characteristic (here, being high-value, or an engagement score) is a **moderator**. Formally we’re estimating the **CATE** — the *Conditional Average Treatment Effect*, $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$: the average email effect *among customers who look like x* . ($Y(1), Y(0)$ are the two potential outcomes — spend if emailed / not — from notebook 01; $\mathbb{E}[\cdot]$ is “average over customers.”) The ATE is just the CATE averaged over everybody.

1.0.2 The tool: a structural model + the do-operator

Notebook 01 used flexible trees (BART) to get the CATE. Here we use a different, more *interpretable* approach — a **structural causal model** via the `pathmc` library. We write down the equation we believe generated spend (including an **interaction term** — a feature that multiplies the treatment, which is exactly how “the effect depends on a moderator” is encoded), fit it, and then ask the model counterfactual questions with the **do-operator**: `do(email=1)` means “*intervene* to email this customer,” as opposed to merely *observing* emailed customers (who differ in other ways). The difference in predicted spend between `do(email=1)` and `do(email=0)`, at a given x , *is* the CATE at x .

The single most important output is the **posterior of the interaction coefficient**: if it’s clearly away from zero, the segments *really* respond differently; if it straddles zero, the “gap” is noise and we shouldn’t act on it. That’s the “is the difference real?” test, answered with uncertainty.

7-step contract: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Customers are **high- or low-value** (`prior_value`) and also carry a continuous **engagement** score. We plant a real, layered moderation: the email lifts low-value customers by **€3** and high-value by **€12** at average engagement, **plus** a smooth **+€8 x engagement** slope on top. So the true per-customer effect is $\tau = 3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ — a genuine, sizeable, and *continuous* heterogeneity. `prior_value` also drives spend directly (must be included); there’s a background trend.

On real data. There’s no single public dataset for this exact setup, but the swap is trivial: your **own CRM / campaign log** already has a segment column (`prior_value`), covariates, a treatment flag (was the customer emailed) and an outcome (`spend`). Point the model at those columns instead of the simulator. The one requirement is that the email was assigned randomly *or* that you’ve measured the confounders (notebook 05) — otherwise the “segment effects” are contaminated by who-got-emailed.

TRUE segment-average effects (at mean engagement): {'low': 7.0, 'high': 16.0}
TRUE effect spans €3.0 ... €19.9 across customers

	email	prior_value	engagement	trend	spend	is_high
0	0.0	low	0.918802	58.466377	23.337758	0.0
1	0.0	low	0.651170	60.223782	19.057907	0.0
2	1.0	high	0.242720	53.530129	50.963256	1.0
3	0.0	low	0.119019	37.977148	12.868787	0.0
4	1.0	low	0.488445	61.807028	29.494100	0.0

1.2 3 · Identify — the CATE as a functional of the model

Before fitting anything, we pin down *exactly* what we’re estimating and whether it’s recoverable.

The estimand. The conditional effect, written with the do-operator (intervene, don’t just observe):

$$\tau(x) = \mathbb{E}[\text{spend} \mid \text{do}(\text{email}=1), x] - \mathbb{E}[\text{spend} \mid \text{do}(\text{email}=0), x].$$

Where heterogeneity lives. We assume `spend` is generated by the structural equation (with $e=\text{email}$, $v=\text{is_high}$, $g=\text{engagement}$)

$$\text{spend} = \beta_0 + \beta_1 e + \beta_2 v + \beta_3 (e \cdot v) + \beta_4 (e \cdot g) + \dots$$

Take the difference between $e=1$ and $e=0$ and everything not multiplied by e cancels, leaving

$$\tau(x) = \beta_1 + \beta_3 v + \beta_4 g.$$

So the email effect is a **baseline** β_1 **plus moderation terms**: **all the heterogeneity is carried by the interaction coefficients** β_3, β_4 . If those were zero, τ would be the same for everyone and `cate()` would collapse to `ate()`. This is why “is the effect heterogeneous?” is literally the question “is β_3 (or β_4) non-zero?” — which we answer with its posterior in Step 5.

Identification. We still need the email->spend link to be unconfounded. Here email was *randomized*, so unconfoundedness holds by design and the required **adjustment set is empty** — the structural model isn’t doing confounding-correction, it’s giving us a clean way to *query subgroup effects as functionals of the posterior*. `pathmc` reads the graph and confirms this:

```
admissible adjustment set(s) for email->spend: [set()]
identifiable: True
```

(An empty adjustment set `[set()]` means “condition on nothing” — correct, because randomization already made email independent of everything. In an *observational* version we’d see the confounders listed here, exactly as in notebook 05.)

1.3 4 · Estimate — fit the structural model

We fit the equation above in one shot; `pathmc` returns a posterior over every coefficient. The `effects_summary()` table below is the direct read-out: each row is a β with its posterior mean and 94% HDI (*Highest Density Interval* — the tightest interval holding 94% of the posterior mass). The two rows to watch are `b_ev` and `b_eg` — the interaction terms. If their intervals sit clearly away from zero, the segments genuinely respond differently; that’s the statistical backing for writing segment-specific rules rather than one blanket policy.

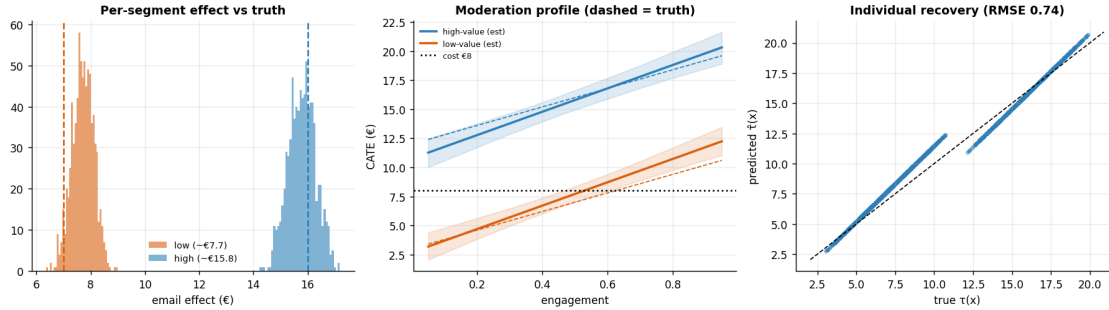
	mean	sd	hdi_3%	hdi_97%
name				
b_e	2.68	0.80	1.20	4.03
b_v	18.41	0.43	17.62	19.24
b_ev	8.08	0.61	6.96	9.24
b_eg	10.05	1.37	7.56	12.52
b_g	3.95	0.94	2.27	5.77
b_tr	0.29	0.02	0.27	0.32

1.4 5 · Validate — recover the effects, the gap, and the moderation profile

Four checks:

1. **Segment CATEs** vs the planted €3 / €12 — with credible intervals.
2. **Is the gap real?** the interaction posterior β_3 ; if it clearly excludes zero, the segments truly differ (not noise).
3. **The continuous moderation profile** — CATE as a smooth function of engagement, recovered vs the true $3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ line. This is the payoff of modelling the continuous moderator rather than crudely binning customers into two groups.
4. **Per-customer recovery** — the effect predicted for each individual customer (from the fitted moderation coefficients) against their true effect, on the 45° line. Recovering *segment averages* is easy; recovering the effect for each *individual* is the real test.

```
ATE (pooled)      €10.80   [10.24147347 11.39268715]
CATE low  (eng .5) € 7.71   (true 7.0)   [6.96124494 8.45150667]
CATE high (eng .5) €15.79   (true 16.0)   [14.90499107 16.78183398]
Interaction beta(email:is_high) mean €8.08, P(>0) = 1.000 -> gap is REAL
```



How to read those three panels. *Left* — the two segment-effect posteriors sit right on top of their planted truths (dashed lines) and barely overlap, so the model both **recovers** the effects and **separates** the segments. *Middle* — the estimated CATE (solid) tracks the true $3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ line (dashed) across the whole engagement range: modelling the *continuous* moderator paid off, because the effect clearly rises with engagement rather than being a flat per-segment number. Note where each line crosses the €8 cost line — that’s the engagement level above which a customer becomes worth emailing. *Right* — predicted vs true effect *per customer* hugs the 45° line (low RMSE), confirming recovery at the individual level, not just on segment averages.

1.5 6 · Decide, in euros — segment rules, and what pooling costs

Now the payoff. First we turn the per-segment effects into concrete **send rules** using the honest probability criterion from notebook 01 — target a segment only where **P(effect > cost)** is **high** (not merely where the mean beats cost). Then we answer the question that justifies this whole analysis: **how much money does segmenting actually make versus treating everyone as the pooled average?**

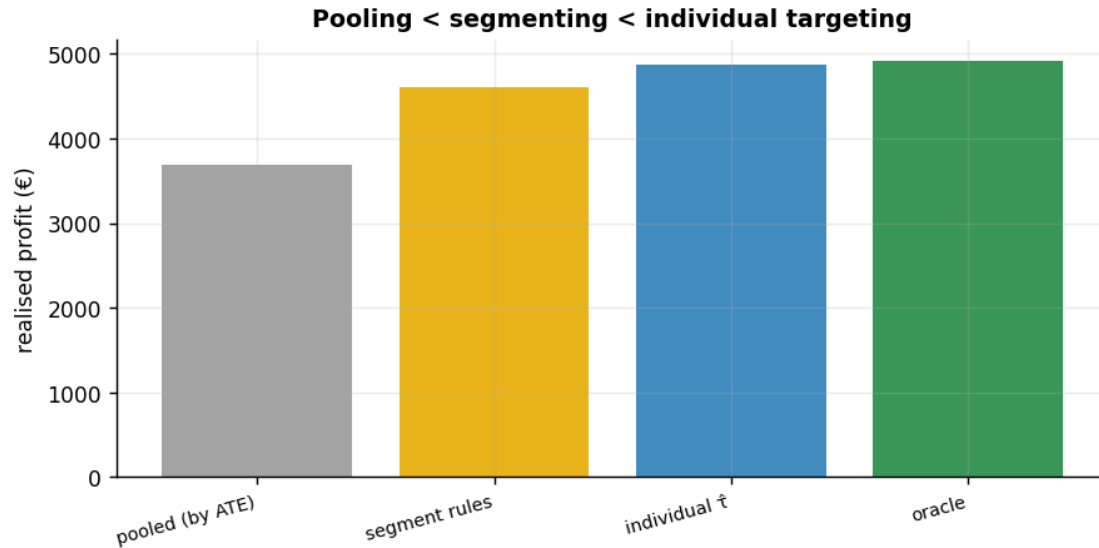
We compare four policies on the *known* per-customer truth (only possible in simulation), forming a ladder of increasing sophistication: **pooled** (one all-or-nothing decision on the ATE) -> **segment rules** (one decision per segment) -> **individual** (a per-customer decision on $\hat{\tau}(x)$) -> **oracle** (the unbeatable rule that knows every true effect). The gaps between the rungs are, literally, the euros that finer targeting is worth.

segment	effect	net_of_cost	P(effect>cost)	action
low-value	7.71	-0.29	0.24	SKIP
high-value	15.79	7.79	1.00	SEND

Realised profit by policy:

pooled (by ATE)	€3,682
segment rules	€4,602
individual tau_hat	€4,871
oracle	€4,923

Moving from pooled to individual targeting is worth €1,189 (32% uplift).



1.6 7 · Caveats

- **Heterogeneity is only as good as the moderator you measured.** If the real driver of differential response isn't `is_high/engagement` (or a proxy), the interaction won't catch it.
- **Interaction != causing the moderator.** We're not claiming *making* someone high-value changes responsiveness — only that responsiveness *differs* across pre-existing segments.
- **Random slopes for many thin segments.** With dozens of small segments, replace the single interaction with partially-pooled random slopes so noisy segments borrow strength (see nb 03).
- **Same unconfoundedness caveat** — here email is randomized; observationally a hidden common cause of email and spend would bias every segment effect (see nb 05).